RM9: Compact local merge state and strong-observer audit

Method-first ramp-metering review

1 October 2026

Status. RM8 Candidate A was approved. This is the first bounded RM9 analytic audit. It derives conditional constraints, not an adopted simulator, a full published-scheme validation, or a useful-gradient certificate. A new finite-receiving allocation decision is pending.

1 Contracts and proof ownership

The task remains one metered finite-storage ramp entering a local multilane mainline. Compare common bounded open-loop rates and count all affected traffic through common clearance. The aggregate observer has the declared road-total density, flow, speed and count histories, ramp histories, known lane parameters, geometry, forcing and prior rates. Observation functions and source equations must be specified before making an exact-history claim.

The conservation structure below is consistent with the lane-model lineage [1]. Projection, identifiability and accounting results are *reviewer derivations* under their stated conditions. They are not reported results of [1–4]. The full update in [1] includes additional modified-demand/supply rules; none is silently repaired or certified here.

2 A conserved sum need not be a closed state

Let lane 1 be the shoulder lane; lanes 2 through m are interior lanes. For a cell of length L, $n_\ell = Lk_\ell$ is mass in vehicle equivalents, $q_\ell^{\text{in/out}}$ and $\lambda_{a\to b}$ are rates in vehicles per unit time. Writing a generic conservative local balance, with realized ramp entry r,

$$\dot{n}_\ell = q_\ell^{\text{in}} - q_\ell^{\text{out}} + \sum_a \lambda_{a\to\ell} - \sum_b \lambda_{\ell\to b} + \mathbf{1}_{\ell=1} r. \tag{1}$$

Define shoulder/rest state $z = (n_1, n_R)$, with $n_R = \sum_{\ell=2}^{m} n_\ell$. Internal transfers among rest lanes cancel. But the boundary transfer $\lambda_{1\to 2}$ depends on lane 2, not generally on n_R alone. For an autonomous exact projected update G, a necessary condition is

$$Px = Px' \implies PF(x,u) = PF(x',u) \quad \text{for every admissible common input } u. \tag{2}$$

Here F is the detailed one-step map and P the proposed projection. If longitudinal/lateral source dependencies, mode memory or receiving states are missing from z, summing balances does not prove this condition.

Explicit local-operator obstruction. For three or more lanes, compare spatially constant profiles in a local plateau

$$(k_1, k_2, k_3) = (a, b, c), \qquad (k_1', k_2', k_3') = (a, b+\epsilon, c-\epsilon), \quad \epsilon \neq 0. \tag{3}$$

Both have the same shoulder/rest masses. On the free-flow branch where the source keep-right incentive is $I = 1 - b/a$, the weighted downstream densities reduce to the constant local densities and the donor demand is $v_1 a$. Choose positive densities with $b, b+\epsilon < a/2$, all sending/receiving limitations inactive. The candidate lateral-demand expression from [1] then reduces to

$$\lambda_{1\to 2}(b) = v_1 a \frac{a-2b}{a+b}, \qquad \frac{d\lambda_{1\to 2}}{db} = -\frac{3v_1 a^2}{(a+b)^2} \neq 0. \tag{4}$$

This is an algebraic source-branch calculation, not a new transfer law. Reverse transfer is inactive on this branch. A homogeneous plateau makes each lane's longitudinal inflow/outflow balance zero locally; rest-internal transfers cancel in n_R. Therefore the projected lateral evolution differs although z agrees. This rules

out a *general autonomous shoulder/rest closure* on that branch without extra information or restrictions. It does **not** rule out a closure conditioned on additional aggregate observations, prove insufficiency of every scalar state, or establish identical observable traffic histories. The two profiles above may have different road flows and speeds.

Two distinct remedies. If the mainline has exactly two physical lanes, "rest" is one lane: there is no within-rest aggregation. With known total $N = n_1 + n_2$, n_1 (or a contrast $n_1 - n_2$) specifies the same lane information. For three or more lanes, retain sufficient receiving-adjacent information, prove a lossless reconstruction from admitted observables, or explicitly adopt and justify an approximate mixing closure. The last option changes the scientific claim and is not approved.

3 Hidden at a snapshot versus hidden from the observer

For a diagnostic equilibrium free-flow observation map, assume each lane has known constant speed v_ℓ, common observation support, no active supply limit, and $q_\ell = v_\ell k_\ell$. Define road-total density K and rate Q:

$$K = \sum_\ell k_\ell, \qquad Q = \sum_\ell v_\ell k_\ell. \tag{5}$$

This condition is a diagnostic subcase, not a declaration that the entire source update always realizes this observation map. For two distinct lane speeds, already K, Q determine the split:

$$k_1 = \frac{Q - v_2 K}{v_1 - v_2}, \qquad k_2 = \frac{v_1 K - Q}{v_1 - v_2}. \tag{6}$$

Thus two different free-flow lane occupations with the same K but different Q do not demonstrate information hidden from this observer. For equal $v_1 = v_2 = v$, the instantaneous flow becomes $Q = vK$ and no longer adds an independent equation. Selecting exact equality solely to obtain this ambiguity would create a symmetry-dependent claim. It is not approved here. Distinct speeds arbitrarily near equality restore algebraic invertibility in the noiseless map; numerical conditioning and noisy estimation are separate questions from exact identity.

Speed has a measurement meaning. A density-weighted space mean is redundant with (5):

$$V_s = Q/K, \qquad K > 0. \tag{7}$$

In the same equilibrium, point-detector subcase, a vehicle-passage-weighted arithmetic speed instead supplies

$$V_t = \frac{\sum_\ell q_\ell v_\ell}{Q} = \frac{\sum_\ell v_\ell^2 k_\ell}{Q}, \qquad Q > 0. \tag{8}$$

For three lanes, distinct known speeds and these co-supported observations,

$$\begin{bmatrix} 1 & 1 & 1 \\ v_1 & v_2 & v_3 \\ v_1^2 & v_2^2 & v_3^2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} K \\ Q \\ QV_t \end{bmatrix}, \qquad \det A = (v_2 - v_1)(v_3 - v_1)(v_3 - v_2) \neq 0. \tag{9}$$

This is an identifiability result for that specific observation map, not a field estimator or a statement about mismatched station/cell averages. A road-average speed must not silently be replaced with Q/K if it means an independent detector statistic. Source [1] distinguishes time/space mean speeds and calibrates distinct lane speeds. Its detector-derived density may not be an independent exact measurement of true cell density; the audit must keep that dependence explicit.

For three lanes with K, Q alone, adding shoulder k_1 determines the remaining pair when $v_2 \neq v_3$. This is a *lossless reparameterization* of three lane densities, not an autonomous two-dimensional group compression. If V_t is also admitted under (9), no additional split information is required in this free-flow diagnostic. Congested modes and other measurement maps require a different proof. Snapshot ambiguity in a higher-dimensional profile still does not establish ambiguity of its full history.

History and memory. The required certificate would compare two realizable trajectories with

$$H(x(t)) = H(x'(t)) \quad \text{at every declared observed time/cell,} \qquad u_{\text{past}}, d, \text{geometry and parameters common.} \tag{10}$$

No trajectory satisfying (10) and differing useful metering responses is constructed in this audit. An unobserved prehistory, changed lane parameters or unequal speed aggregates cannot be concealed in such a pair. Likewise, a deterministic hysteresis mode updated from admitted density history and common initial mode [3] is reconstructible memory, not an extra hidden state. The field mechanism [4] does not by itself settle this identifiability issue.

4 Meter permission is not guaranteed physical entry

Let u be meter-permitted rate; d_r the local ramp sending offer at the merge, affected by meter releases, ramp propagation and stored traffic; d_s the shoulder-mainline offer; r, f_s their realized entry rates; S finite receiving rate. At a common receiving point necessary constraints include

$$0 \le r \le d_r, \qquad 0 \le f_s \le d_s, \qquad r + f_s \le S. \tag{11}$$

An upstream meter constrains its own release $e_r(t) \le u(t)$, not necessarily instantaneous merge entry $r(t)$: released traffic can arrive later. Only an explicitly co-located offer gate gives $d_r(t) \le u(t)$. That special mapping is not adopted here. Account for post-meter in-flight mass if it exists.

At the full multilane interface, every incoming longitudinal/lateral transfer must share its target receiving budget; every outgoing transfer must also respect its donor budget. Equation (11) isolates a two-input component; it does not authorize reserving supply twice or prescribe an update order.

With two offers exceeding shared supply, these inequalities alone admit many allocations. Conservation and a finite capacity do not select the ramp/mainline service split. A consistent priority/redistribution rule is needed before there is a unique forward map, let alone its derivative. The ramp-control source [2] prioritizes ramp flow and excludes mainline spillback onto ramps. Its assumption cannot replace a finite-access contract for arbitrary demand. Finite-receiving ramp preference itself remains a valid alternative when its bounds and domain are made explicit.

The primary node framework [5] describes fixed-priority allocation and unused supply redistribution. It cautions against setting priorities proportional to current demand because of nodal invariance. Therefore the neutral proportional *batch* allocation used in the separate VSL analytical construction cannot be inherited as a universally sound continuous-flow merge law. No change to that closed VSL result follows: the traffic semantics and scope differ.

5 Complete accounting rejects a queue-location-only benefit

Let $M(t)$ be all uncompleted affected traffic inside the accounting boundary, including ramp and external waiting, mainline cell masses and residual demand. For common total arrivals $A(t)$ and departure rate $E(t)$,

$$\dot{M}(t) = A(t) - E(t), \qquad J = \int_0^{T_c} M(t)\,dt, \qquad M(T_c) = 0. \tag{12}$$

J is total system time, with a common sufficiently long clearance horizon T_c; a delay objective may subtract a policy-independent reference time. If two policies have common initial total mass, arrivals and full departure history, they have identical $M(t)$ and J. Moving waiting from the ramp to the mainline alone cannot improve this objective.

On a fixed differentiable branch with control parameter θ, common forcing and common terminal accounting, the sensitivity is

$$\frac{dJ}{d\theta} = -\int_0^{T_c} (T_c - t)\frac{\partial E(t;\theta)}{\partial \theta}\,dt. \tag{13}$$

This follows by integrating the sensitivity of (12) and exchanging the order of integration. It is not an independent gradient verification. If a work-conserving fixed-supply bottleneck remains continuously busy with unchanged aggregate departures for all compared perturbations, the derivative is zero even if local queues differ. A useful rate decision must change actual departure timing, future service/spillback, or another explicitly approved outcome. Merely introducing finite shared receiving does not ensure it.

6 Disposition and next gate

The compact family is not refuted. However, autonomous group closure is not inherited, lane contrast need not be hidden from aggregate flow/speed, and queue redistribution alone does not establish an all-stream action benefit. The complete source scheme has unresolved branch/flow-budget transcription issues, while the ramp extension uses a priority assumption outside the general finite-access contract. No equation repair or symmetry assumption is adopted.

RM9-UG1 asks for a finite-merge allocation contract: start with source-backed demand-independent priorities plus unused-supply redistribution (recommended), use explicitly bounded ramp preference, or adopt a deliberately stylized proportional-cohort law with a narrower claim. After that decision, continue a source-to-conservative-formulation audit and test exact information matching before attempting an action/gradient certificate. Priorities' numerical values, spatial discretization and lane count are not frozen by this report. All configurations must preserve the strong observer. No experiment is required or authorized by this gate.

References

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[2] Nagalur Subraveti, Knoop, van Arem. *Lane Change Control Combined with Ramp Metering*. 2021. 10.1155/2021/8638523. RM0004, Sections 2.1–3. Final repository PDF inspected.

[3] Cao, Nilsson, Coogan. *On the Impact of the Capacity Drop Phenomenon for Freeway Traffic Flow Control*. 2021. arXiv:2106.10347. RM0003, Section II.B, equation 3; archived PDF and public methods.

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